

Multiple Choice

1. **Answer:** B

Options: A) To measure interstellar mass.; B) To measure galactic distances.; C) To measure galactic energy-density.; D) To measure interstellar density.

Note: Correct option: B - To measure galactic distances.

2. **Answer:** D

Options: A) a dark ring visible from Earth composed of dark dusty particles; B) the most opaque ring of Saturn made of highly reflective ice particles; C) the widest ring of Saturn located between two large ring gaps; D) a large gap visible from Earth produced by an orbital resonance with the moon Mimas

Note: Correct option: D - a large gap visible from Earth produced by an orbital resonance with the moon Mimas

3. **Answer:** D

Options: A) the Sun mainly emits blue light.; B) the atmosphere absorbs mostly blue light.; C) molecules scatter red light more effectively than blue light.; D) molecules scatter blue light more effectively than red light.

Note: Correct option: D - molecules scatter blue light more effectively than red light.

4. **Answer:** A

Options: A) Distance at which one astronomical unit measures one arcsecond from Earth.; B) Orbital distance of the solar system around the center of the Milky Way in one year.; C) Effective distance of the solar wind (i.e. the radius of the heliosphere).; D) Historical distance to the brightest star Sirius.

Note: Correct option: A - Distance at which one astronomical unit measures one arcsecond from Earth.

5. **Answer:** A

Options: A) 750 million years.; B) 2 billion years.; C) 3 billion years.; D) 4.5 billion years.

Note: Correct option: A - 750 million years.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'A and B only'.
10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Dione'.

Long Form Response

11. **Answer:** 250 years. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** They are leftover planetesimals that never accreted into planets.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.